Vectors in n Dimensions

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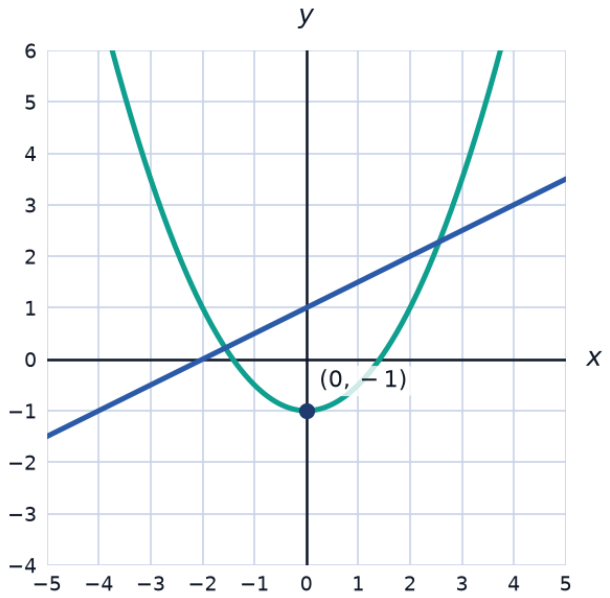
Part I — Free-Response Questions (FRQs): Foundational Fluency (1–7)

1 [Foundational] • **Target: Vector Arithmetic and Linear Combinations**

Consider two vectors $\mathbf{v}_1 = (3; 4)$ and $\mathbf{v}_2 = (-4; 3)$. What is the dot product $\mathbf{v}_1 \cdot \mathbf{v}_2$?

2 [Foundational] • **Target: Linear Independence and Dependence**

Consider the vectors $\mathbf{v}_1 = (3; 4)$ and $\mathbf{v}_2 = (-4; 3)$. What is the dot product $\mathbf{v}_1 \cdot \mathbf{v}_2$?

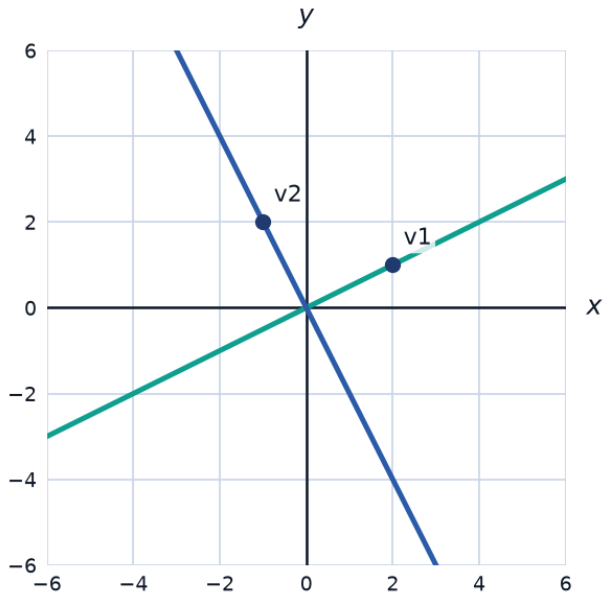


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3 [Foundational] • Target: Dot Product, Norm, and Orthogonality

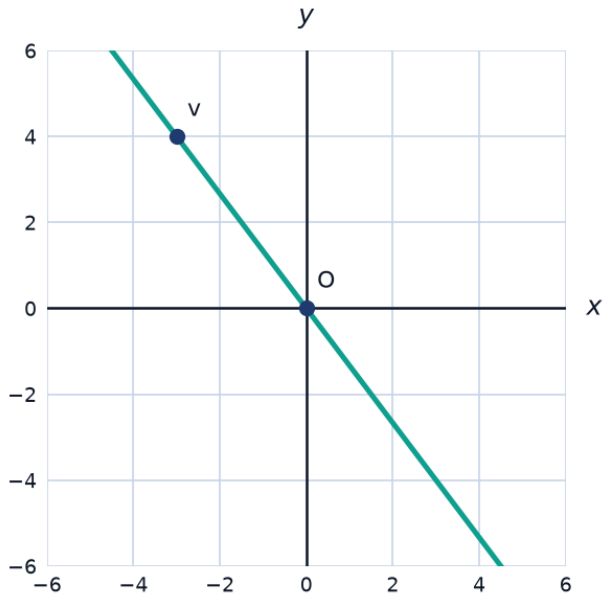
Express the complete solution set of the non-homogeneous linear system in parametric vector form:

$\{x_1 + 2x_2 - x_3 + 3x_4 = 4; 2x_1 + 4x_2 - x_3 + 8x_4 = 11; -x_1 - 2x_2 + 2x_3 - x_4 = -1$



4 [Foundational] • Target: Orthogonal Projections

Consider the vector $\mathbf{v} = (-3; 4)$ in $\mathbb{R}^2$. What is the magnitude (or norm) of this vector?



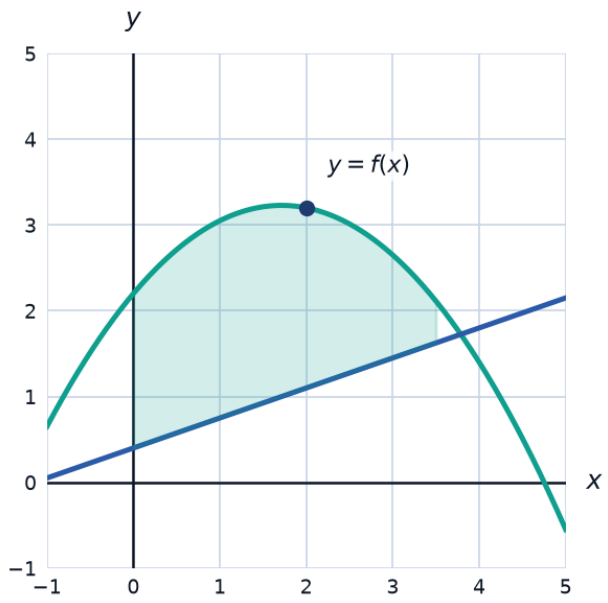
5 [Foundational] • Target: Vector Arithmetic and Linear Combinations

Consider the vector $\mathbf{a} = (2\ k)$ and the vector $\mathbf{b} = (-1\ 3)$. \\ Part 1: If $\mathbf{a}$ and $\mathbf{b}$ are orthogonal, find the value of k. \\ Part 2: Let the resulting vector from Part 1 be $\mathbf{v}$. The transformation T is defined by the matrix $M = (1, 2\ 2, -1)$. \\ If $T(\mathbf{v}) = (x\ y)$, find the value of x.

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6 [Foundational] • Target: Linear Independence and Dependence

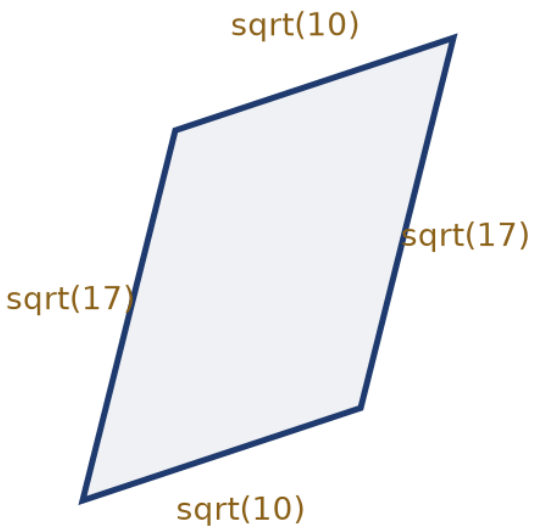
A linear transformation $T : \mathbb{R}^2 \to \mathbb{R}^2$ is defined by $T(x, y) = (ax + by, cx + dy)$. The matrix representation of T with respect to the standard basis is $A = (a, b\ c, d)$. The image of the standard basis vectors $\mathbf{e}_1 = (1, 0)$ and $\mathbf{e}_2 = (0, 1)$ under T are given by $T(\mathbf{e}_1) = (3, -1)$ and $T(\mathbf{e}_2) = (1, 2)$. The transformation is applied to a square with vertices $(0, 0)$, $(1, 0)$, $(0, 1)$, $(1, 1)$. The area of the transformed square is known to be k. Find the value of k.



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7 [Foundational] • Target: Dot Product, Norm, and Orthogonality

A parallelogram in the xy-plane is defined by the vectors $\mathbf{u} = (3; 1)$ and $\mathbf{v} = (1; 4)$. What is the area of this parallelogram?

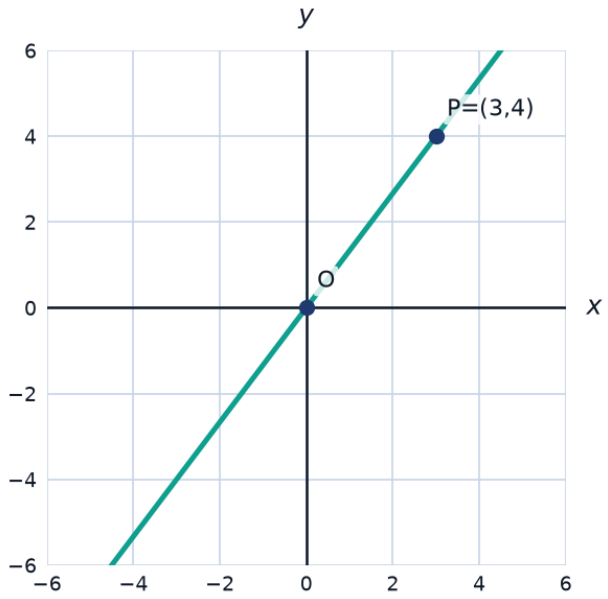


Part II — Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

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8 [Intermediate] • Target: Orthogonal Projections

Consider the vector $\mathbf{v} = (3; 4)$. If this vector is plotted on a standard xy-plane, it represents a directed line segment from the origin to the point $(3, 4)$. What is the geometric interpretation of the length of this vector?

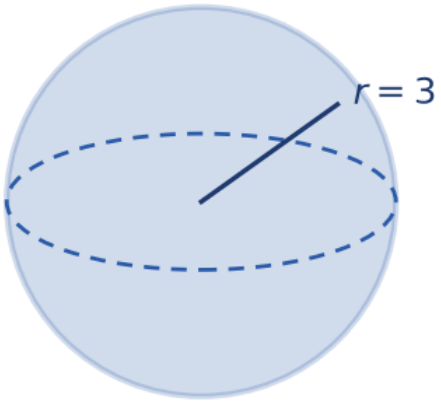


9 [Intermediate] • Target: Vector Arithmetic and Linear Combinations

Consider the parallelepiped in $\mathbb{R}^3$ defined by the vectors $\mathbf{u} = (2, 1, -1)$, $\mathbf{v} = (1, k, 2)$, and $\mathbf{w} = (3, 4, 1)$. The volume of this parallelepiped is given as 15 cubic units. Find the possible value(s) of k.

10 [Intermediate] • Target: Linear Independence and Dependence

Consider a parallelepiped in $\mathbb{R}^3$ formed by the origin and the three vectors $\mathbf{u} = (2\ 1\ k)$, $\mathbf{v} = (1\ 3\ 0)$, and $\mathbf{w} = (0\ 2\ 4)$. The volume of this parallelepiped is given as 24 cubic units. Find the value of k if $k > 0$.



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11 [Intermediate] • Target: Dot Product, Norm, and Orthogonality

Let $T : \mathbb{R}^2 \to \mathbb{R}^2$ be a linear transformation represented by the matrix $A = (3, -1\ 1, 2)$ with respect to the standard basis $\{\mathbf{e}_1, \mathbf{e}_2\}$. Consider a new basis $B = \{\mathbf{v}_1, \mathbf{v}_2\}$ where $\mathbf{v}_1 = (1\ 1)$ and $\mathbf{v}_2 = (1\ -1)$. Let P be the change-of-basis matrix from basis B to the standard basis. Find the matrix representation of T with respect to basis B, denoted by $[T]_B$. This matrix can be expressed as $P^{-1}AP$. Find the sum of the diagonal entries of $[T]_B$.

12 [Intermediate] • Target: Orthogonal Projections

Solve the initial value problem for the coupled system of linear differential equations:
$\mathbf{x}'(t) = (1, 4; 2, 3)\mathbf{x}(t), \qquad \mathbf{x}(0) = (5; 1).$

13 [Intermediate] • Target: Vector Arithmetic and Linear Combinations

Consider a linear transformation $T : \mathbb{R}^3 \to \mathbb{R}^2$ defined by $T(x, y, z) = (2x - y + z, x + 3y - 2z)$. Let V be the image of T, i.e., $V = \text{Im}(T)$. Furthermore, let W be the subspace of $\mathbb{R}^2$ defined by $W = \{(a, b) \in \mathbb{R}^2 \mid a + b = 0\}$. Find the dimension of the intersection of V and W, i.e., $\dim(V \cap W)$.

14 [Intermediate] • Target: Linear Independence and Dependence

Prove the Parallelogram Law for arbitrary vectors $\mathbf{u}, \mathbf{v} \in \mathbb{R}^n$: $||\mathbf{u} + \mathbf{v}||^2 + ||\mathbf{u} - \mathbf{v}||^2 = 2||\mathbf{u}||^2 + 2||\mathbf{v}||^2$. What is its geometric significance?

Part III — Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)

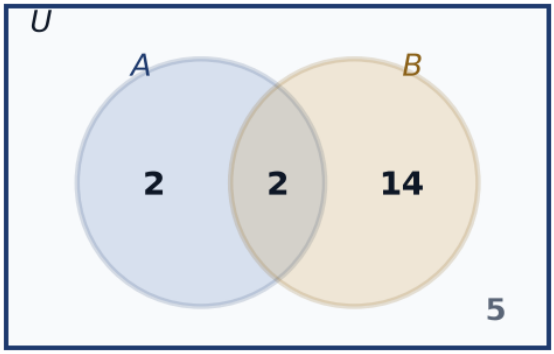
15 [Challenging] • Target: Dot Product, Norm, and Orthogonality • Cross-Lesson: 4.01 Determinants

The vector $\mathbf{v} = (-2; 1)$ is an eigenvector of the matrix $A = (3, 2; 1, 4)$. What is the corresponding eigenvalue λ?

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16 [Challenging] • Target: Orthogonal Projections • Cross-Lesson: 2.01 Systems of Linear Equations

Let $M_2(\mathbb{R})$ be the vector space of 2×2 real matrices. Determine, with formal verification via the Subspace Test, whether $W = \{A \in M_2(\mathbb{R}) : A^T = -A\}$ (the set of skew-symmetric matrices) is a subspace of $M_2(\mathbb{R})$. Find a basis and $\dim(W)$.



17 [Challenging] • Target: Vector Arithmetic and Linear Combinations • Cross-Lesson: 3.01 Matrices & Matrix Algebra

Express the invertible matrix $A = (1, 2; 3, 4)$ as a product of elementary matrices.

18 [Challenging] • Target: Linear Independence and Dependence • Cross-Lesson: 4.01 Determinants

Consider the following system of linear equations represented by the augmented matrix: $(1, 2, |, 5; 2, -1, |, 0)$. Use row reduction to transform the matrix into row-echelon form and determine the unique solution (x, y) for the system. What is the value of x?

19 [Challenging] • Target: Dot Product, Norm, and Orthogonality • Cross-Lesson: 2.01 Systems of Linear Equations

Let $\mathbf{u} = (1, -2, 2, 0)^T$ and $\mathbf{v} = (2, 1, 0, -2)^T$ in $\mathbb{R}^4$. Compute $||\mathbf{u}||$, $||\mathbf{v}||$, the inner product $\mathbf{u} \cdot \mathbf{v}$, the cosine of the angle θ between them, and verify the Cauchy–Schwarz inequality.

20 [Challenging] • Target: Orthogonal Projections • Cross-Lesson: 3.01 Matrices & Matrix Algebra

Consider the linear transformation $T : \mathbb{R}^2 \to \mathbb{R}^2$ represented by the matrix $A = (3, 1; 1, 3)$. Identify one of the eigenvalues of A.

Part IV — Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

21 [Foundational] • Target: Vector Arithmetic and Linear Combinations

Given $\mathbf{u} = [2, -1, 4]$ and $\mathbf{v} = [-3, 5, 2]$ in $\mathbb{R}^3$, compute $3\mathbf{u} - 2\mathbf{v}$.

a. $[12, -13, 8]$

b. $[0, 7, 16]$

c. $[12, -7, 8]$

d. $[-12, 13, -8]$

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22 [Foundational] • Target: Vector Arithmetic and Linear Combinations

Is the vector $\mathbf{w} = [7, 4]$ a linear combination of $\mathbf{v}_1 = [1, 2]$ and $\mathbf{v}_2 = [3, -1]$?

a. Yes, $\mathbf{w} = \frac{19}{7}\mathbf{v}_1 + \frac{10}{7}\mathbf{v}_2$

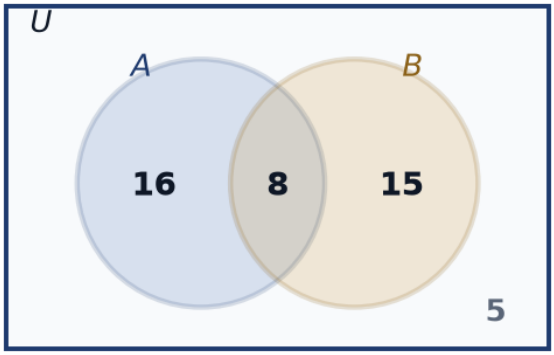
b. No, the vectors are linearly dependent

c. Yes, $\mathbf{w} = 2\mathbf{v}_1 + \mathbf{v}_2$

d. No, because the determinant is zero

23 [Foundational] • Target: Vector Arithmetic and Linear Combinations

What is the definition of the span of a set of vectors $\{\mathbf{v}_1, \ldots, \mathbf{v}_k\}$ in $\mathbb{R}^n$?



a. The set of all possible linear combinations $c_1\mathbf{v}_1 + \ldots + c_k\mathbf{v}_k$ where $c_i \in \mathbb{R}$

b. The set of all vectors perpendicular to $\mathbf{v}_1$

c. The sum of the lengths of the vectors

d. The matrix formed by using the vectors as columns

24 [Foundational] • Target: Linear Independence and Dependence

A set of vectors $\{\mathbf{v}_1, \ldots, \mathbf{v}_k\}$ is linearly independent if and only if:

a. $c_1\mathbf{v}_1 + \ldots + c_k\mathbf{v}_k = \mathbf{0}$ implies $c_1 = c_2 = \ldots = c_k = 0$

b. At least one vector is a multiple of another

c. The dot product of any two vectors is zero

d. The number of vectors k is greater than the dimension n

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25 [Foundational] • Target: Linear Independence and Dependence

Can any set of 4 vectors in $\mathbb{R}^3$ be linearly independent?

a. No, any set of more than n vectors in $\mathbb{R}^n$ is guaranteed to be linearly dependent

b. Yes, provided none of the vectors is the zero vector

c. Yes, if they all have unit length

d. Only if their cross products are non-zero

26 [Foundational] • Target: Linear Independence and Dependence

Are the vectors $\mathbf{v}_1 = [1, 2, 0]$, $\mathbf{v}_2 = [0, 1, 3]$, and $\mathbf{v}_3 = [2, 5, 3]$ linearly independent?

a. No, because $\mathbf{v}_3 = 2\mathbf{v}_1 + \mathbf{v}_2$

b. Yes, they span all of $\mathbb{R}^3$

c. No, because they contain a zero entry

d. Yes, because their determinant is 1

27 [Foundational] • Target: Dot Product, Norm, and Orthogonality

Compute the Euclidean norm (length) of the vector $\mathbf{v} = [2, -3, 6]$ in $\mathbb{R}^3$.

a. 7

b. $\sqrt{11}$

c. 49

d. $\sqrt{41}$

28 [Foundational] • Target: Dot Product, Norm, and Orthogonality

Find the unit vector in the direction of $\mathbf{u} = [-1, 2, 2]$.

a. $\left[-\frac{1}{3}, \frac{2}{3}, \frac{2}{3}\right]$

b. $[-1, 2, 2]$

c. $\left[-\frac{1}{9}, \frac{2}{9}, \frac{2}{9}\right]$

d. $\left[-\frac{1}{\sqrt{5}}, \frac{2}{\sqrt{5}}, 0\right]$

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Part V — Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

29 **[Intermediate]** • **Target: Dot Product, Norm, and Orthogonality**

What does the Cauchy-Schwarz inequality state for any vectors $\mathbf{u}, \mathbf{v} \in \mathbb{R}^n$?

a. $|\mathbf{u} \cdot \mathbf{v}| \leq ||\mathbf{u}||||\mathbf{v}||$

b. $||\mathbf{u} + \mathbf{v}|| \leq ||\mathbf{u}|| + ||\mathbf{v}||$

c. $\mathbf{u} \cdot \mathbf{v} = ||\mathbf{u}||||\mathbf{v}||$

d. $|\mathbf{u} \cdot \mathbf{v}| \geq ||\mathbf{u}||||\mathbf{v}||$

30 **[Intermediate]** • **Target: Orthogonal Projections**

What is the formula for the orthogonal projection of vector $\mathbf{y}$ onto a non-zero vector $\mathbf{u}$?

a. $\text{proj}_{\mathbf{u}}(\mathbf{y}) = \frac{\mathbf{y} \cdot \mathbf{u}}{\mathbf{u} \cdot \mathbf{u}}\mathbf{u}$

b. $\text{proj}_{\mathbf{u}}(\mathbf{y}) = \frac{\mathbf{y} \cdot \mathbf{u}}{||\mathbf{y}||}\mathbf{u}$

c. $\text{proj}_{\mathbf{u}}(\mathbf{y}) = (\mathbf{y} \cdot \mathbf{u})\mathbf{u}$

d. $\text{proj}_{\mathbf{u}}(\mathbf{y}) = \mathbf{y} - \mathbf{u}$

31 **[Intermediate]** • **Target: Orthogonal Projections**

Compute the projection of $\mathbf{y} = [7, 6]$ onto $\mathbf{u} = [4, 2]$.

a. $[8, 4]$

b. $[4, 2]$

c. $[7, 6]$

d. $[2, 1]$

32 **[Intermediate]** • **Target: Orthogonal Projections**

If $\hat{\mathbf{y}} = \text{proj}_{\mathbf{u}}(\mathbf{y})$, what is the geometric relationship between $\mathbf{z} = \mathbf{y} - \hat{\mathbf{y}}$ and $\mathbf{u}$?

a. $\mathbf{z}$ is orthogonal to $\mathbf{u}$

b. $\mathbf{z}$ is parallel to $\mathbf{u}$

c. $\mathbf{z}$ has unit length

d. $\mathbf{z} = \mathbf{0}$ always

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33 [Intermediate] • Target: Vector Arithmetic and Linear Combinations

Consider the vector space $\mathbb{R}^2$ with the standard dot product. Let $\mathbf{u} = (2, 3)$ and $\mathbf{v} = (-3, 2)$. Which of the following statements about $\mathbf{u}$ and $\mathbf{v}$ is true?

a. $\mathbf{u}$ and $\mathbf{v}$ are orthogonal.

b. The angle between $\mathbf{u}$ and $\mathbf{v}$ is acute.

c. The angle between $\mathbf{u}$ and $\mathbf{v}$ is obtuse.

d. $\mathbf{u}$ and $\mathbf{v}$ are linearly dependent.

34 [Intermediate] • Target: Linear Independence and Dependence

Consider the matrix $A = (1, 1; 1, -1)$. Let $\mathbf{v} = (3; 1)$. Suppose we want to find the vector $\mathbf{x}$ of the form $c_1\mathbf{a}_1 + c_2\mathbf{a}_2$ where $\{\mathbf{a}_1, \mathbf{a}_2\}$ is an orthogonal basis derived from the columns of A, such that $||\mathbf{v} - \mathbf{x}||$ is minimized. Find the value of $||\mathbf{x}||$.

a. $\sqrt{10}$

b. $\sqrt{20}$

c. $\sqrt{5}$

d. $2\sqrt{10}$

35 [Intermediate] • Target: Dot Product, Norm, and Orthogonality

Consider the linear transformation $T : \mathbb{R}^3 \to \mathbb{R}^3$ defined by $T(\mathbf{x}) = A\mathbf{x}$, where $A = (1, 2, 3; 4, 5, 6; 7, 8, 9)$. Let V be the image of T, i.e., $V = \text{Im}(T)$. V is a subspace of $\mathbb{R}^3$. Find the dimension of the orthogonal complement of V, denoted by $V^\perp$.

a. 1

b. 2

c. 0

d. 3

36 [Intermediate] • Target: Orthogonal Projections

Consider a vector $\mathbf{v} = (3\ k)$ in $\mathbb{R}^2$. If the magnitude of $\mathbf{v}$ is $\sqrt{13}$, find the value of k given that $k > 0$.

a. 1

b. 2

c. 3

d. 4

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37 [Intermediate] • Target: Vector Arithmetic and Linear Combinations

Consider the vector space $\mathbb{R}^2$ with the standard dot product. Let $\mathbf{u} = (3, 4)$ and $\mathbf{v} = (2, -1)$. Calculate the dot product $\mathbf{u} \cdot \mathbf{v}$.

a. 10

b. 2

c. -2

d. 6

Part VI — Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)

38 [Challenging] • Target: Linear Independence and Dependence • Cross-Lesson: 4.01 Determinants

Consider the vectors $\mathbf{u} = (2, -3)$ and $\mathbf{v} = (4, 1)$ in $\mathbb{R}^2$. What is the dot product $\mathbf{u} \cdot \mathbf{v}$?

a. 5

b. 8

c. 4

d. 11

39 [Challenging] • Target: Dot Product, Norm, and Orthogonality • Cross-Lesson: 2.01 Systems of Linear Equations

Consider the matrix $A = (1, 2; -2, k)$. If the column vectors of A are orthogonal, find the value of k.

a. 2

b. -1

c. 4

d. -2

40 [Challenging] • Target: Orthogonal Projections • Cross-Lesson: 3.01 Matrices & Matrix Algebra

Which of the following is a subspace?

a. x

b. $\{p \in P_2 : p(0) = 1\}$

c. x

d. $\{A \in M_{2\times 2} : \det A = 0\}$

41 [Challenging] • Target: Vector Arithmetic and Linear Combinations • Cross-Lesson: 4.01 Determinants

Consider the vector $\mathbf{v} = (5; -12)$. Calculate the magnitude of $\mathbf{v}$, denoted as x.

a. 13

b. 17

c. 7

d. 169

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42 [Challenging] • Target: Linear Independence and Dependence • Cross-Lesson: 2.01 Systems of Linear Equations

Which of the following is NOT a subspace?

a. x

b. $\{A \in M_{2\times 2} : A^T = A\}$

c. x

d. x

43 [Challenging] • Target: Dot Product, Norm, and Orthogonality • Cross-Lesson: 3.01 Matrices & Matrix Algebra

Consider the vector $\mathbf{v} = (5 \ \ -12)$. Calculate the magnitude of $\mathbf{v}$, denoted as x.

a. 13

b. 7

c. 169

d. 17

44 [Challenging] • Target: Orthogonal Projections • Cross-Lesson: 4.01 Determinants

For which k is $\langle k, -\frac{1}{2}, \frac{3}{2}\rangle$ in $\text{span}\{\langle -1, 1, -3\rangle\}$?

a. $\frac{3}{2}$

b. $\frac{1}{2}$

c. $-\frac{1}{2}$

d. -1

45 [Challenging] • Target: Vector Arithmetic and Linear Combinations • Cross-Lesson: 2.01 Systems of Linear Equations

Let B be the basis $\langle -1, -1, 1\rangle, \langle -1, -2, 2\rangle, \langle -2, 1, -2\rangle$ of R^3. Find $[v]\,B$ for $\mathbf{v} = \langle 8, 3, -1\rangle$.

a. $\langle 8, 3, -1\rangle$

b. $\langle 2, 1, 9\rangle$

c. $\langle -3, -1, -2\rangle$

d. $\langle 3, 1, 2\rangle$

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